

```
Last login: Mon Jun 22 15:09:02 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % cd /Users/adityadave/PE/PE-Daywise-Work-Tracker/19-JUNE/Q2\&Q21
adityadave@Adityas-MacBook-Air-3 Q2&Q21 % python3 script.py
Running 1/5...
^CTraceback (most recent call last):
  File "/Users/adityadave/PE/PE-Daywise-Work-Tracker/19-JUNE/Q2&Q21/script.py", line 69, in <module>
    output, _ = proc.communicate(QUERY)
  File "/Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/subprocess.py", line 1134, in communicate
    stdout, stderr = self._communicate(input, endtime, timeout)
  File "/Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/subprocess.py", line 1979, in _communicate
    ready = selector.select(timeout)
  File "/Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/selectors.py", line 416, in select
    fd_event_list = self._selector.poll(timeout)
KeyboardInterrupt
```

```
adityadave@Adityas-MacBook-Air-3 Q2&Q21 % python3 script.py
```

```
Running 1/5...
```

```
Running 2/5...
```

```
Running 3/5...
```

```
Running 4/5...
```

```
Running 5/5...
```

```
=====
STH RUN OUTPUT
=====
```

```
SET
```

```
WARNING: Log 1: Planner Input:
```

```
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTTRY
:alias <>
:eref
  {ALIAS
:aliasname supplier
:colnames ("s_suppkkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:reldid 17308
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
  {BOOLEXP
:boolop and
:args (
  {OPEXP
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 6
:varatype 1700
:var typmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
  {CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTTRY
:alias
  {ALIAS
:aliasname n
:colnames <>
}
:eref
  {ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_commen
```

```

    t")
  }
:rtkind 0
:relid 17285
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
  {ALIAS
:aliasname r
:colnames <>
}
:eref
  {ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtkind 0
:relid 17291
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperinfos {
  {RTEPERMISSIONINFO
:relid 17285
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
  {RTEPERMISSIONINFO
:relid 17291
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
}
:jointree
  {FROMEXPR
:fromlist (
  {RANGETBLREF
:rtindex 1
}
  {RANGETBLREF
:rtindex 2
}
)
:quals
  {BOOLEXP
:boolop and
:args (
  {OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
  {VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
  {OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
  {VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
  {OPEXP
:opno 1054
:opfuncid 1048
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
:location -1
}
:location -1
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
:location -1
}
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname l1
:colnames <>
}
:eref
{ALIAS
:aliasname l1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17349
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
  {BOOLEXP
    :boolop and
    :args (
      {SUBLINK
        :sublinkType 0
        :subLinkId 0
        :testexpr <>
        :operName <>
        :subselect
        {QUERY
          :commandType 1
          :querySource 0
          :canSetTag true
          :utilityStmt <>
          :resultRelation 0
          :hasAggs false
          :hasWindowFuncs false
          :hasTargetSRFs false
          :hasSubLinks false
          :hasDistinctOn false
          :hasRecursive false
          :hasModifyingCTE false
          :hasForUpdate false
          :hasRowSecurity false
          :hasGroupRTE false
          :isReturn false
          :ctelList <>
        }
        :rtable (
          {RANGETBENTRY
            :alias
            {ALIAS
              :aliasname o
              :colnames <>
            }
          }
        )
      }
    )
  }
)
:aliasname o
}

```

```

        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_to
talprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_sh
ippriority" "o_comment")
    }
    :rtekind 0
    :relid 17337
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromC1 true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17337
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
):jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
  :quals
  {BOOLEXP
    :boolop and
    :args (
      {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
          {VAR
            :varno 2
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 1
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 1
            :location -1
          }
        )
      }
      :location -1
    }
    {SCALARARRAYOPEXP
      :opno 1054
      :opfuncid 1048
      :hashfuncid 0
      :negfuncid 0
      :useOr true
      :inputcollid 100
      :args (
        {VAR
          :varno 1
          :varattno 6
          :vartype 1042
          :vartypmod 19
          :varcollid 100
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 6
          :location -1
        }
        {ARRAYEXP
          :array_typeid 1014
          :array_collid 100
          :element_typeid 1042
          :elements (
            {CONST
              :consttype 1042
              :consttypmod -1
              :constcollid 100
              :constlen -1
              :constbyval false
              :constisnull false
              :location -1
              :constvalue 12 [ 48 0 0 0 49 45 85 82 71 69 78 8
4 ]
            }
            {CONST
              :consttype 1042
              :consttypmod -1
              :constcollid 100
              :constlen -1
              :constbyval false
              :constisnull false
              :location -1
              :constvalue 10 [ 40 0 0 0 50 45 72 73 71 72 ]
            }
          )
        }
        :multidims false
        :list_start -1
        :list_end -1
        :location -1
      )
    }
    :location -1
  }
)
:location -1
}
):mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
    }
  }
)

```

```

        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:CommandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_su
pplycost" "ps_comment")
}
:rekind 0
:relid 17318
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17318
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
)
}
:location -1
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false

```

```

:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
  {VAR
    :varno 2
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 3
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 521
:opfncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
  }
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
      }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
)
{RANGETBENTRY
:alias <>
:eref
  {ALIAS
    :aliasname orders
    :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
  }
:rtekind 0
:relid 17337
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
  {ALIAS
    :aliasname nation
    :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
  }
}

```

```

    }
    :rtekind 0
    :releid 17285
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 4
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
  :alias <>
  :eref
    {ALIAS
    :aliasname *GROUP*
    :colnames (*s_name*)
    }
  :rtekind 9
  :groupexprs (
    {VAR
    :varno 1
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 51
    }
  )
  :lateral false
  :inFromCl false
  :securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
  :releid 17308
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9 11)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :releid 17349
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 10 19 20)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :releid 17337
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 10)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :releid 17285
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
  {RANGETBLREF
  :rtindex 2
  }
  {RANGETBLREF
  :rtindex 3
  }
  {RANGETBLREF
  :rtindex 4
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 176
    }
    {VAR
    :varno 2
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 3
    :location 188
    }
  )
  :location 186
  }
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 213
}
{VAR
:varno 2
:varattno 1
:varatype 23
:varattpmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 226
}
)
:location 224
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 3
:varattno 3
:varatype 1042
:varattpmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 252
}
{CONST
:consttype 1042
:consttppmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 268
:constvalue 5 [ 20 0 0 0 70 ]
}
)
:location 266
}
{OPEXPR
:opno 1097
:opfuncid 1089
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 13
:varatype 1082
:varattpmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 13
:location 284
}
{VAR
:varno 2
:varattno 12
:varatype 1082
:varattpmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 12
:location 303
}
)
:location 301
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:opetName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname 12
:colnames <>
}
:eref
{ALIAS
:aliasname 12
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 0
:relid 17349
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{BOOLEXPR
:boolop and
:args (

```

```

{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStat <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orders
tatus" "o_totalprice" "o_orderdate" "o_orderp
riority" "o_clerk" "o_shippriority" "o_commen
t")
}
:rtekind 0
:relid 17337
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17337
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:location -1
}
)
{SCALARARRAYOPEXP
:opno 1054
:opfncid 1048
:hashfncid 0
:negfncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
{ARRAYEXP
:array_typeid 1014
:array_collid 100
:element_typeid 1042
:elements (
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 12 [ 48 0 0 0 49 45 85 8
2 71 69 78 84 ]
}
)
}
)
}
)
}

```

```

        :consttype 1042
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 50 45 72 7
        3 71 72 ]
    }
    )
    :multidims false
    :list_start -1
    :list_end -1
    :location -1
    }
    )
    :location -1
    }
    )
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
        {TARGETENTRY
        :expr
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
        }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
    }
    {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
    {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSublinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :cteList <>
    :rtable (
        {RANGETBLENTY
        :alias
        {ALIAS
        :aliasname ps
        :colnames <>
        }
        :eref
        {ALIAS
        :aliasname ps
        :colnames ("ps_partkey" "ps_suppkey" "ps_avai
        lqty" "ps_supplycost" "ps_comment")
        }
        :rtekind 0
        :relid 17318
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
        }
    )
    :rteperminfos (
        {RTEPERMISSIONINFO
        :relid 17318
        :inh true
        :requiredPerms 2
        :checkAsUser 0
        :selectedCols (b 8 9 10)
        :insertedCols (b)
        :updatedCols (b)
        }
    )
    }
    :jointree
    {FROMEXPR
    :fromlist (
        {RANGETBLREF
        :rtindex 1
        }
    )
    :quals
    {BOOLEXP
    :boolop and
    :args (
        {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0

```

```

:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
)
:location -1
}
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
  }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>

```

```

        :withCheckOptions <>
        :stmt_location -1
        :stmt_len -1
    }
    :location -1
}
)
:location -1
}
)
}
)
:rtperminfos (
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 420
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 436
}
}
:location 434
}
{OPEXP
:opno 518
:opfuncid 144
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 472
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 488
}
}
:location 485
}
)
:location 468
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 363
}
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 363
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 363
}
:resno 3
:resname 1_supplekey
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 363
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 363
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 363
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 363
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 363
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR

```

```

:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 363
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 363
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 363
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 363
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 363
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 363
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 363
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 17349

```

```

:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 363
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 16
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 331
}
{BOOLEXP
:boolop not
:args (
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname l3
:colnames <>
}
:eref
{ALIAS
:aliasname l3
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_lin
enumbr" "l_quantity" "l_extendedprice" "l_discount" "
l_tax" "l_returnflag" "l_linestatus" "l_shipdate" "l_c
ommitdate" "l_receiptdate" "l_shipinstruct" "l_shipmod
e" "l_comment")
}
:rtekind 0
:relid 17349
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{BOOLEXP
:boolop and
:args (
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_ord
erstatus" "o_totalprice" "o_orderdate" "o_
orderpriority" "o_clerk" "o_shippriority"
"o_comment")
}
:rtekind 0
:relid 17337
:inh true

```

```

:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
)
:rttperminfos (
{RTEPERMISSIONINFO
:relid 17337
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGEtblref
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:location -1
}
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{ARRAYEXPR
:array_typeid 1014
:array_collid 100
:element_typeid 1042
:elements (
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 12 [ 48 0 0 0 49 45 8
5 82 71 69 70 84 ]
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 50 45 7
2 73 71 72 ]
}
}
:multidims false
:list_start -1
:list_end -1
:location -1
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0

```

```

:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_a
vailqty" "ps_supplycost" "ps_comment")
}
:rtekind 0
:reldid 17318
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reldid 17318
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2

```

```

        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location -1
    }
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    }
    :location -1
    }
    {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 ]
    }
    )
    :location -1
    }
    }
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 1 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDecls <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :location -1
    }
    }
    :location -1
    }
    }
    }
    :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 17349
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21
    22 23)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    :quals
    {BOOLEXPR
    :boolop and
    :args (
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    }
    )
    }
    }

```

```

:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 616
  }
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location 632
  }
)
:location 630
}
{OPEXPR
:opno 518
:opfuncid 144
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location 668
  }
  {VAR
    :varno 2
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 3
    :location 684
  }
)
:location 681
}
{OPEXPR
:opno 1097
:opfuncid 1089
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 13
    :location 719
  }
  {VAR
    :varno 1
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 12
    :location 738
  }
)
:location 736
}
:location 664
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 559
    }
  }
  :resno 1
  :resname l_orderkey
  :ressortgroupref 0
  :resorigtbl 17349
  :resorigcol 1
  :resjunk false
}
{TARGETENTRY
:expr
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 559
  }
}

```

```

    }
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 559
}
}
:resno 3
:resname l_suppkey
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 559
}
}
:resno 4
:resname l_linenumber
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 559
}
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 559
}
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 559
}
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 559
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 559
}
:resno 9
:resname 1_returnflag
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 559
}
:resno 10
:resname 1_linestatus
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 559
}
:resno 11
:resname 1_shipdate
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 559
}
:resno 12
:resname 1_commitdate
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 559
}
:resno 13
:resname 1_receiptdate
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 559
}
:resno 14
:resname 1_shipinstruct
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 559
}
:resno 15
:resname 1_shipmode
:ressortgroupref 0
:resorigtbl 17349
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr

```

```

        {VAR
        :varno 1
        :varattno 16
        :vartype 1043
        :vartypmod 48
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 16
        :location 559
        }
        :resno 16
        :resname l_comment
        :ressortgroupref 0
        :resorigtbl 17349
        :resorigcol 16
        :resjunk false
        }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
    }
    :location 527
    }
)
:location 523
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 776
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 790
}
)
:location 788
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 814
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 823
:constvalue 13 [ 52 0 0 0 65 82 71 69 78 84 73 78 65 ]
}
)
:location 821
}
)
:location 209
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
}
:resno 1
:resname s_name
:ressortgroupref 2
:resorigtbl 17308
:resorigcol 2
:resjunk false

```

```

    }
    {TARGETENTRY
    :expr
      {AGGREF
      :aggfnoid 2803
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes <>
      :aggdirectargs <>
      :args <>
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggsplit 0
      :aggno -1
      :aggtransno -1
      :location 67
      }
      :resno 2
      :resname numwait
      :ressortgroupref 1
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
      }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause (
      {SORTGROUPCLAUSE
      :tleSortGroupRef 2
      :eqop 1054
      :sortop 1058
      :reverse_sort false
      :nulls_first false
      :hashable true
      }
    )
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause (
      {SORTGROUPCLAUSE
      :tleSortGroupRef 1
      :eqop 410
      :sortop 413
      :reverse_sort true
      :nulls_first true
      :hashable true
      }
      {SORTGROUPCLAUSE
      :tleSortGroupRef 2
      :eqop 1054
      :sortop 1058
      :reverse_sort false
      :nulls_first false
      :hashable true
      }
    )
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDecls <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
  }

WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 propparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:table (
{RANGETBLENTY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17285
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 17291
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)

```

```

:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 17285
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 0 10)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :relid 17291
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 0 9)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
  {RANGETBLREF
  :rtindex 2
  }
)
:quals (
  {OEXPR
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
    }
    {PARAM
    :paramkind 1
    :paramid 0
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
    }
  )
  :location -1
  }
  {OEXPR
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
  )
  :location -1
  }
  {OEXPR
  :opno 1054
  :opfuncid 1040
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 100
  :args (
    {VAR
    :varno 2
    :varattno 2
    :vartype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
    }
    {CONST
    :consttype 1042
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
  )
  :location -1
  }
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>

```

```

:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_sel elec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=0 in safe_list=0 > reason=outer_reference

WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_sel elec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_sel elec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_sel elec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = <>

WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStat <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:reldid 17285
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:reldid 17291
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
)
:teperinfos (
{RTEPERMISSIONINFO
:reldid 17285
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reldid 17291
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}

```

```

{RANGETBLREF
:rtindex 2
}
):quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
}
):mergeActionList <>
):mergeTargetRelation 0
):mergeJoinCondition <>
):targetList (
{TARGETENTRY
:expr (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
):override 0
):onConflict <>
):returningOldAlias <>
):returningNewAlias <>
):returningList <>
):groupClause <>
):groupDistinct false
):groupingSets <>
):havingQual <>
):windowClause <>
):distinctClause <>
):sortClause <>
):limitOffset <>
):limitCount <>
):limitOption 0
):rowMarks <>
):setOperations <>
):constraintDeps <>
):withCheckOptions <>
):stmt_location -1
):stmt_len -1
}
)
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause

```

```

<> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 426 2097) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 1 40 0 0 69 85 82 79 80 69 }} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b 1 2) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.0058823529411764705 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:ctelist <>
:table (
{RANGEABLEENTRY
:alias
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtckind 0
:reld 17337
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:teperminfos (
{RTEPERMISSIONINFO
:reld 17337
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 2
:paramtype 23
:vartypmod -1
:paramcollid 0
:location -1
}
}
:location -1
}
)
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0
0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0
]
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:smt_location -1
:smt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({(VAR :varno 1 :varattno 6 :var
type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -
1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1
) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} (RESTRIC
TINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlev
elsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_jo
in false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :
outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVAL
ENCMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_
is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcoll
id 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buck
etsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=2 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({(VAR :varno 1 :varattno 6 :va
rtype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -
1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b
1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} (RESTRIC
TINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varle
velsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_j
oin false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b)
:outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVA
LENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_
is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcol
lid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buc
ketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:table {
{RANGETBLENTRY
:alias
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o
_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 17337
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17337
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtdindex 1
}
)
}
:quals (
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048

```

```

:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 1042
    :vartypmod 19
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {CONST
    :consttype 1014
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0
      0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]
  }
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :resortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :var
type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -
1 :constbyval false :constisnull false :location -1 :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1
) :incompatible_relids (b) :outer_relids (b) :left_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :va
rtype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen
-1 :constbyval false :constisnull false :location -1 :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b
1) :incompatible_relids (b) :outer_relids (b) :left_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | paral
lel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:cteList <>
:rtable (
  {RANGESTLENTY
    :alias
    {ALIAS
      :aliasname ps
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname ps
      :colnames ("ps_partkey" "ps_supplekey" "ps_availqty" "ps_supplycost" "p
s_comment")
    }
    :rtekind 0
    :relid 17318
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
  }
)
}

```

```

:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17318
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 4
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 5
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>

```

```

:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 521 :opfncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :
vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true
:constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volat
ile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_co
st :startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_m
cvmfreq -1 :right_mvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} (RESTRICTINFO :clause (OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :v
arno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 4 :paramtype 23 :pa
ramtypmod -1 :paramcollid 0 :location -1}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :n
um_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.p
er_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup
0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEME
MBER :em_expr (PARAM :paramkind 1 :paramid 4 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_re
lds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvmfreq -1 :right_mvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} (RESTRICTINFO :clause (O
PEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varret
urningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1) :is_pushed_down true :can_join false :pseud
oconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b
1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_
expr (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) :em_relds (b 1) :em_is_const false
:em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr (PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location
-1) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left
_mvmfreq -1 :right_mvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=4 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 521 :opfncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23
:vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval tru
e :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volat
ile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_c
ost :startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left
_mvmfreq -1 :right_mvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} (RESTRICTINFO :clause (OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :
varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 4 :paramtype 23 :p
aramtypmod -1 :paramcollid 0 :location -1}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :
num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.
per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup
0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEME
MBER :em_expr (PARAM :paramkind 1 :paramid 4 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_r
elds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvmfreq -1 :right_mvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} (RESTRICTINFO :clause :
OPEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varre
turningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1) :is_pushed_down true :can_join false :pseu
doconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (
b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_
expr (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) :em_relds (b 1) :em_is_const false
:em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr (PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location
-1) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :lef
t_mvmfreq -1 :right_mvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2: After pull up At starting of Path Generation:
DETAIL:
  QUERY
  :
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :ctelist <>
  :table (
    {RANGETBLENTTRY
      :alias
      :aliases
      :aliasname ps
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname ps
      :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
    }
    :rtekind 0
    :reld 17318
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  )
  )
  :rteperminfos (
    {RTEPERMISSIONINFO
      :reld 17318
      :inh true
      :requiredPerms 2
      :checkAsUser 0
      :selectedCols (b 8 9 10)
      :insertedCols (b)
      :updatedCols (b)
    }
  )
  :jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
  }
  :quals (
    {OPEXPR
      :opno 521
      :opfncid 147
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        (VAR
          :varno 1
          :varattno 3
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 3
          :location -1
        )
        {CONST
          :consttype 23
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location -1
          :constvalue 4 [ 0 0 0 0 0 0 0 ]
        }
      )
      :location -1
    }
  )
  )
  )

```

```

:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location -1
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:smt_location -1
:smt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 521 :opfncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 521 :opfncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:rtelist <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname o
      :colnames <>
    }
  }
  :eref
  {ALIAS
    :aliasname o
    :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
  }
  :rtekind 0
  :relid 17337
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
)
}
:rtperminfos (
  {RTEPERMISSIONINFO
    :relid 17337
    :inh true
    :requiredPerms 2
    :checkAsUser 0
  }
)

```

```

:selectCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
rtindex 1
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 8
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen
:constbyval false
:constisnull false
:location -1
:constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 }}
:location -1
}
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:smt_location -1
:smt_len -1
}
}

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :var type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :left_mcvmfreq -1 :right_mcvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICT INFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 8 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :right_em (EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 8 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvmfreq -1 :right_mcvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED

WARNING: max_parallel_hazard_walker: Param paramid=8 in_safe_list=0 → reason=outer reference

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :var type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvmfreq -1 :right_mcvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICT INFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 8 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :right_em (EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 8 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvmfreq -1 :right_mcvmfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:

DETAIL: QUERY

```

:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0

```

```

:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname o
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname o
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 0
    :relid 17337
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17337
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals (
      {SCALARARRAYOEXPR
        :opno 1054
        :opfuncid 1048
        :hashfuncid 0
        :negfuncid 0
        :useOr true
        :inputcollid 100
        :args (
          {VAR
            :varno 1
            :varattno 6
            :vartype 1042
            :vartypmod 19
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 6
            :location -1
          }
          {CONST
            :consttype 1014
            :consttypmod -1
            :constcollid 100
            :constlen -1
            :constbyval false
            :constisnull false
            :location -1
            :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0
              0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0
            ]
          }
        )
        :location -1
      }
    )
  }
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels 1 :required_rels (b 1

```

) :incompatible_relids (b) :outer_relids (b) :left_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args {(VAR :varno 1 :varattno 6 :va
rtype 1042 :vartypemod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} (CONST :consttype 1014 :consttypemod -1 :constcollid 100 :constlen
-1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b
1) :incompatible_relids (b) :outer_relids (b) :left_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | paral
lel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:teList <>
:table (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 17318
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17318
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypemod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 10
:paramtype 23
:paramtypemod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypemod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 11
:paramtype 23
:paramtypemod -1
:paramcollid 0
:location -1
}
)
:location -1
}
}

```

```
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
  }
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 10 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :location -1} :em_relds (b 1)} :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 10 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {PARAM :paramkind 1 :paramid 11 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b 1) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :location -1} :em_relds (b 1)} :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 11 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b 1)} :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED

WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:

DETAIL: (QUERY

```
commandType 1
querySource 0
canSetTag true
utilityStmt <>
resultRelation 0
hasAggs false
hasWindowFuncs false
hasTargetSRFs false
hasSubLinks false
hasDistinctOn false
hasRecursive false
hasModifyingCTE false
hasForUpdate false
hasRowSecurity false
hasGroupRE false
isReturn false
codelist <>
:table (
  {RANGEBLENTY
    :alias
    {ALIAS
      :aliasname ps
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname ps
      :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
    }
    :rtekind 0
    :relid 17318
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
  }
```

```

:inFromCl true
:securityQuals <>
}
)
:rtoperminfos (
{RTEPERMISSIONINFO
:relid 17318
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
){CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:groupingsets
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvfreq -1 :right_mvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvfreq -1 :right_mvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE

```

```

WARNING: is_parallel_safe: INPUT qual = (<{VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = (<{VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname o
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname o
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 0
    :relid 17337
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17337
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
      {PARAM
        :paramkind 1
        :paramid 14
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
  }
  :location -1
}
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 1042
    :vartypmod 19
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
  }
  {CONST
    :consttype 1014
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0
0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0
]
}
)
}
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>

```

```
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ((VAR :varno 1 :varattno 6 :var
type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -
1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1)
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1
) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} (RESTRICT
INFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlev
elsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 14 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1}) :is_pushed_down true :can_j
oin false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b)
:outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVA
LENCMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em
_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}) :right_em {EQUIVALENCMEMBER :em_expr {PARAM :paramkind 1 :paramid 14 :paramtype 23 :paramtypmod -1 :paramco
llid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bu
cketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=14 in safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ((VAR :varno 1 :varattno 6 :va
rtype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen
-1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1)
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b
1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} (RESTRIC
TINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlev
elsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) {PARAM :paramkind 1 :paramid 14 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1}) :is_pushed_down true :can_j
oin false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b)
:outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIV
ALENCMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :e
m_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}) :right_em {PARAM :paramkind 1 :paramid 14 :paramtype 23 :paramtypmod -1 :paramc
ollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_b
ucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctlist <>
:table (
{RANGEBLENTTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtckind 0
:reld 17337
:inh false
:relkind r
:reilockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:teperminfos (
{RTEPERMISSIONINFO
:reld 17337
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGEBLENTREF
:rtindex 1
}
)
:quals (
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
){CONST
:consttype 1014
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]
}
)
}
```

```

    )
    :location -1
  }
}
):mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
):override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :var
type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -
1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1})
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1
) :incompatible_relids (b) :outer_relids (b) :left_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 1 :varattno 6 :va
rtypmod 19 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen
-1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]}) :location -1})
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b
1) :incompatible_relids (b) :outer_relids (b) :left_relids (b) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | paral
lel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowfuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTF false
:isReturn false
:cteList <>
:table (
  {RANGETBLENTY
    :alias
    {ALIAS
      :aliasname ps
      :colnames <>
    }
  }
  :eref
  {ALIAS
    :aliasname ps
    :colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
  }
  :rtekind 0
  :relid 17318
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
)
):
):rteperinfos (
  {RTEPERMISSIONINFO
    :relid 17318
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 9 10)
    :insertedCols (b)
    :updatedCols (b)
  }
)
):jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals (
      {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false

```

```

:opcollid 0
:inputcollid 0
:args (
  (VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  )
  (PARAM
    :paramkind 1
    :paramid 16
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  )
)
:location -1
)
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  (VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  )
  (PARAM
    :paramkind 1
    :paramid 17
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  )
)
:location -1
)
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  (VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  )
  (CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
  )
)
:location -1
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile :security_level 0 :num_base_rels 1 :clause_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}) :location -1} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {PARAM :paramkind 1 :paramid 17 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}) :location -1} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} {PARAM :paramkind 1 :paramid 17 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=16 in_safe_list=0 => reason=outer_reference

WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

```

ost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relids (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relids (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relids (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relids (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 2 :location -1} {PARAM :paramkind 1 :paramid 17 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} :em_relids (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relids (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 17 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:table (
:table {
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p_s_comment")
}
:rtekind 0
:reldid 17318
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reldid 17318
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rteindex 1
}
)
:quals (
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
){CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 521 :opfncid 147 :opresultype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 { 0 0 0 0 0 0 0 0 }}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b 1) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 521 :opfncid 147 :opresultype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 { 0 0 0 0 0 0 0 0 }}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b 1) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:relList <>
:table (
{RANGELEFTENTRY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtckind 0
:relid 17308
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{OEXPR
:opno 1756
:opfncid 1720
:opresultype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 { 24 0 0 0 0 -128 }
}
)
:location -1
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 1
:plan_name SubPlan\ 1

```

```

:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
)
:startup_cost 0
:per_call_cost 24.2625
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramids (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 24.4875
:per_call_cost 0.0025
}
)
}
))
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname l1
:colnames <>
}
}
:eref
{ALIAS
:aliasname l1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17349
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 3
:plan_name SubPlan\ 3
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 2)
:args (
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
)
}
:startup_cost 0
:per_call_cost 14646
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0

```

```

:args (
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 3
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramids (i 3)
:plan_id 4
:plan_name SubPlan\ 4
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 13970.6625
:per_call_cost 0.0025
}
)
}
{ALTERNATIVESUBPLAN
:subplans (
  {SUBPLAN
    :subLinkType 0
    :testexpr <>
    :paramids <>
    :plan_id 5
    :plan_name SubPlan\ 5
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse true
    :parallel_safe false
    :setParam <>
    :parParam (i 4 5)
    :args (
      {VAR
        :varno 2
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno 2
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 3
        :location -1
      }
    )
    :startup_cost 0
    :per_call_cost 9472
  }
  {SUBPLAN
    :subLinkType 2
    :testexpr
    {BOOLEXPR
      :boolop and
      :args (
        {OPEXPR
          :opno 96
          :opfuncid 65
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            {VAR
              :varno 2
              :varattno 2
              :vartype 23
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 2
              :varattnosyn 2
              :location -1
            }
            {PARAM
              :paramkind 1
              :paramid 6
              :paramtype 23
              :paramtypmod -1
              :paramcollid 0
              :location -1
            }
          )
        }
        :location -1
      }
      {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 2
            :varattno 3
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 3
            :location -1
          }
        )
      }
    )
  }
)
}

```

```
(PARAM
:paramkind 1
:paramid 7
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
)
}
:location -1
}
}
:location -1
}
:paramsids (i 6 7)
:plan_id 6
:plan_name SubPlan\ 6
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 8871.94
:per_call_cost 0.005
}
)
}
))
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 17337
:inh false
:relkind r
:relockmode 1
:perminfoindex 3
:tablesampl <>
:lateral false
:inFromC1 true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17285
:inh false
:relkind r
:relockmode 1
:perminfoindex 4
:tablesampl <>
:lateral false
:inFromC1 true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("s_name")
}
}
:rtekind 9
:groupexprs (
(VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varrattnosyn 2
:location 51
}
)
:lateral false
:inFromC1 false
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname l2
:colnames <>
}
}
:eref
{ALIAS
:aliasname l2
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenummer" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 0
:relid 17349
:inh false
:relkind r
:relockmode 1
:perminfoindex 5
:tablesampl <>
:lateral false
:inFromC1 true
:securityQuals ((
(ALTERNATIVESUBPLAN
:subplans (
(SUBPLAN
:sublinkType 0
:testexpr <>
:paramsids <>
:plan_id 7
:plan_name SubPlan\ 7
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 8)
:args (
(VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6

```

```

        :varattnosyn 1
        :location -1
    }
}
:startup_cost 0
:per_call_cost 14646
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 9
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 9)
:plan_id 8
:plan_name SubPlan\ 8
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 13970.6625
:per_call_cost 0.0025
}
)
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 9
:plan_name SubPlan\ 9
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 10 11)
:args (
{VAR
:varno 6
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location -1
}
{VAR
:varno 6
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 3
:location -1
}
)
:startup_cost 0
:per_call_cost 9472
}
{SUBPLAN
:subLinkType 2
:testexpr
{BOOLEXP
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location -1
}
}
{PARAM
:paramkind 1
:paramid 12
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 6
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 6
    :varattnosyn 3
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 13
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
)
:location -1
}
}
:paramids (i 12 13)
:plan_id 10
:plan_name SubPlan\ 10
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 8871.94
:per_call_cost 0.005
}
)
}
)
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname 13
:colnames <>
}
:eref
{ALIAS
:aliasname 13
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17349
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 11
:plan_name SubPlan\ 11
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 14)
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location -1
}
)
:startup_cost 0
:per_call_cost 14646
}
{SUBPLAN
:subLinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location -1
}
)
:location -1
}
}
:paramids (i 15)
:plan_id 12
:plan_name SubPlan\ 12
:firstColType 23
:firstColTypmod -1

```

```

:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 13970.6625
:per_call_cost 0.0025
}
)
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramIds <>
:plan_id 13
:plan_name SubPlan\ 13
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 16 17)
:args (
{VAR
:varno 7
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 2
:location -1
}
{VAR
:varno 7
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location -1
}
)
:startup_cost 0
:per_call_cost 9472
}
{SUBPLAN
:sublinkType 2
:testexpr
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 2
:location -1
}
}
{PARAM
:paramkind 1
:paramid 18
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location -1
}
}
{PARAM
:paramkind 1
:paramid 19
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
)
:location -1
}
)
:paramIds (i 18 19)
:plan_id 14
:plan_name SubPlan\ 14
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 8871.94
:per_call_cost 0.005
}
)

```

```

    }
  })
}
:rtperminfos (
{RTEPERMISSIONINFO
:relid 17308
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 19 20)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17337
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17285
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
}
):jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 5
:isNatural false
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
):quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 176
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 188
}
)
:location 186
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 213
}
{VAR
:varno 2
:varattno 1

```

```

:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 226
}
)
:location 224
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 3
:varattno 3
:varstype 1042
:varstypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 252
}
{CONST
:consttype 1042
:conststypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 268
:constvalue 5 [ 20 0 0 0 70 ]
}
)
:location 266
}
{OPEXPR
:opno 1097
:opfuncid 1009
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 13
:varstype 1002
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 13
:location 284
}
{VAR
:varno 2
:varattno 12
:varstype 1002
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 12
:location 303
}
)
:location 301
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 776
}
{VAR
:varno 4
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 790
}
)
:location 788
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 4
:varattno 2
:varstype 1042
:varstypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 814
}
{CONST
:consttype 1042
:conststypmod -1
:constcollid 100
:constlen -1
:constbyval false

```

```

        :constisnull false
        :location 823
        :constvalue 13 [ 52 0 0 0 65 82 71 69 78 84 73 78 65 ]
    }
}
:location 821
}
}
}
:rarg
{RANGETBLREF
:rtindex 6
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location 420
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 436
}
}
:location 434
}
{OPEXPR
:opno 518
:opfuncid 144
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 3
:location 472
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 488
}
}
:location 485
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 7
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location 616
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 632
}
}
)
:location 630
}
{OPEXPR
:opno 518
:opfuncid 144
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7

```

```

:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location 668
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 684
}
}
:location 681
}
{OPEXPR
:opno 1097
:opfuncid 1099
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 13
:location 719
}
{VAR
:varno 7
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 12
:location 738
}
}
:location 736
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 1
:resname s_name
:ressortgroupref 2
:resorigtbl 17308
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 67
}
:resno 2
:resname numwait
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>

```

```

--distinctClause <
--sortClause <
{
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 410
    :sortpos 13
    :reverse_sort true
    :nulls_first true
    :hashable true
  }
}
{
  {SORTGROUPCLAUSE
    :tleSortGroupRef 2
    :eqop 106
    :sortpos 1858
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
}
}
--limitOffset <
--limitCount <
--limitOption 0
--withCheckOptions <
--stmt_location 0
--stmt_len 0
}
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1756 :opfncmd 1720 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :vartattno 6 :vartype 17 :varlevelsup 0 :varlevelsups 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 1 :varattosyn 6 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constnbval false :constnsullm false :location -1 :constval 6 [ 2 4 0 0 0 -128 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause (ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramsId <> :plan_id 1 :plan_name SubPlan1 :firstColType 2278 :firstColTypmod -1 :firstCollCollection 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParameter <> :paramParam (i 0) :args ((VAR :varno 1 :vartattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 1 :varattosyn 4 :location -1) :startup_cost 0 :per_call_cost 24.2625) (SUBPLAN :sublinkType 2 :testexpr <> :paramsId <> :plan_id 1 :plan_name SubPlan1 :firstColType 23 :firstColTypmod -1 :firstCollCollection 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParameter <> :paramParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025))) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) :mergofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = false | reason = PROPAGATE_RESTRICTED

WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1756 :opfncmd 1720 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :vartattno 6 :vartype 17 :varlevelsup 0 :varlevelsups 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 1 :varattosyn 6 :location -1) (CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constnbval false :constnsullm false :location -1 :constval 6 [ 2 4 0 0 0 -128 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause (ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramsId <> :plan_id 1 :plan_name SubPlan1 :firstColType 2278 :firstColTypmod -1 :firstCollCollection 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParameter <> :paramParam (i 0) :args ((VAR :varno 1 :vartattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 1 :varattosyn 4 :location -1) :startup_cost 0 :per_call_cost 24.2625) (SUBPLAN :sublinkType 2 :testexpr <> :paramsId <> :plan_id 5 :plan_name SubPlan1 :firstColType 2278 :firstColTypmod -1 :firstCollCollection 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParameter <> :paramParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025))) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) :mergofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = false | reason = PROPAGATE_RESTRICTED

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramsId <> :plan_id 3 :plan_name SubPlan1 :firstColType 2278 :firstColTypmod -1 :firstCollCollection 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParameter <> :paramParam (i 2) :args ((VAR :varno 2 :vartattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 2 :varattosyn 1 :location -1) :startup_cost 0 :per_call_cost 14466) (SUBPLAN :sublinkType 2 :testexpr (OPEXPR :opno 96 :opfncmd 65 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :vartattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 2 :varattosyn 2 :location -1) (PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :paramsId (i 1) :plan_id 2 :plan_name SubPlan1 :firstColType 23 :firstColTypmod -1 :firstCollCollection 0 :useHashTable false :unknownEqFalse true :parallel_safe true :setParameter <> :paramParam <> :args <> :startup_cost 24.4875 :per_call_cost 0.0025))) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) :mergofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = false | reason = PROPAGATE_RESTRICTED

WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramsId <> :plan_id 3 :plan_name SubPlan1 :firstColType 2278 :firstColTypmod -1 :firstCollCollection 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParameter <> :paramParam (i 2) :args ((VAR :varno 2 :vartattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 2 :varattosyn 1 :location -1) :startup_cost 0 :per_call_cost 14466) (SUBPLAN :sublinkType 2 :testexpr (OPEXPR :opno 96 :opfncmd 65 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :vartattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 2 :varattosyn 2 :location -1) (PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :paramsId (i 3) :plan_id 4 :plan_name SubPlan1 :firstColType 23 :firstColTypmod -1 :firstCollCollection 0 :useHashTable false :unknownEqFalse true :parallel_safe true :setParameter <> :paramParam <> :args <> :startup_cost 13970.6625 :per_call_cost 0.0025))) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) :mergofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = false | reason = PROPAGATE_RESTRICTED

WARNING: max_parallel_hazard_walker: SubPlan plan_id=3 parallel_safe=> reason=correlated_subplan

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1854 :opfncmd 1848 :presulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 3 :vartattno 3 :vartype 17 :varlevelsup 0 :varlevelsups 0 :varnullingrels (b) :varlevelsups 0 :varreturningtype 0 :varnosyn 3 :varattosyn 3 :location -1) (CONST :consttype 1842 :consttypmod -1 :constcollid 100 :constlen -1 :constnbval false :constnsullm false :location -1 :constval 5 [ 2 0 0 0 7 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b 3) :left_relds (b 3) :right_relds (b 3) :orclause <> :rinfo_serial 7 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) :mergofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = false | reason = PROPAGATE_RESTRICTED

WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1854 :opfncmd 1848 :presulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 3 :vartattno 3 :vartype 17 :varlevelsup 0 :varlevelsups 0 :varnullingrels (b) :varlevelsups 0 :var
```

QUERY PLAN

```
Sort (cost=130254146433.64..130254146433.75 rows=43 width=34) (actual time=1241.938..1241.953 rows=0.00 loops=1)
Output: supplier.s_name, (count(*))
Sort Key: (count(*)) DESC, supplier.s_name
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=2431 read=52525
-> GroupAggregate (cost=130254146431.72..130254146432.47 rows=43 width=34) (actual time=1241.914..1241.928 rows=0.00 loops=1)
Output: supplier.s_name, count(*)
Group Key: supplier.s_name
Buffers: shared hit=2425 read=52525
-> Sort (cost=130254146431.72..130254146431.83 rows=43 width=26) (actual time=1241.912..1241.926 rows=0.00 loops=1)
Output: supplier.s_name
Sort Key: supplier.s_name
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=2425 read=52525
-> Nested Loop Semi Join (cost=0.00..130254146430.55 rows=43 width=26) (actual time=1241.905..1241.918 rows=0.00 loops=1)
Output: supplier.s_name
Join Filter: ((i2.l.suppkey <> i1.l.suppkey) AND (orders.o_orderkey = i2.l_orderkey))
Buffers: shared hit=2425 read=52525
-> Nested Loop (cost=0.00..86836156439.96 rows=58 width=38) (actual time=1241.905..1241.915 rows=0.00 loops=1)
Output: supplier.s_name, i1.l.suppkey, i1.l_orderkey, orders.o_orderkey
Join Filter: ((i3.l.suppkey <> i1.l.suppkey) AND (i3.l_orderkey = i1.l_orderkey))
```

```
Buffers: shared hit=2425 read=52525
-> Nested Loop (cost=0.00..43418315368.47 rows=107 width=38) (actual time=1241.899..1241.902 rows=0.00 loops=1)
  Output: supplier.s_name, l1.l_supplekey, l1.l_orderkey, orders.o_orderkey
  Join Filter: ((l1.l_orderkey = orders.o_orderkey)
  Buffers: shared hit=2425 read=52525
  -> Seq Scan on public.orders (cost=0.00..13521.00 rows=218085 width=4) (actual time=1.076..57.292 rows=218715.00 loops=1)
    Output: orders.o_orderkey, orders.o_custkey, orders.o_orderstatus, orders.o_totalprice, orders.o_orderdate, orders.o_orderpriority, orders.o_clerk, orders.o_shipper
    iority, orders.o_comment
    Filter: (orders.o_orderstatus = 'F'::bpchar)
    Rows Removed by Filter: 231285
    Buffers: shared hit=847 read=7949
  -> Materialize (cost=0.00..43417578896.25 rows=221 width=34) (actual time=0.005..0.005 rows=0.00 loops=218715)
    Output: supplier.s_name, l1.l_supplekey, l1.l_orderkey
    Storage: Memory Maximum Storage: 17kB
    Buffers: shared hit=1578 read=45476
  -> Nested Loop (cost=0.00..43417578896.14 rows=221 width=34) (actual time=1160.335..1160.338 rows=0.00 loops=1)
    Output: supplier.s_name, l1.l_supplekey, l1.l_orderkey
    Join Filter: ((supplier.s_supplekey = l1.l_supplekey)
    Buffers: shared hit=1578 read=45476
    -> Seq Scan on public.lineitem l1 (cost=0.00..43417496980.61 rows=150017 width=8) (actual time=215.530..1107.573 rows=456835.00 loops=1)
      Output: l1.l_orderkey, l1.l_partkey, l1.l_supplekey, l1.l_linenumbr, l1.l_quantity, l1.l_extendedprice, l1.l_discount, l1.l_tax, l1.l_returnflag, l1.l_inestatus, l1.l_shipdate, l1.l_commitdate, l1.l_receiptdate, l1.l_shipmode, l1.l_comment
      Filter: (((l1.l_receiptdate > l1.l_commitdate) AND (ANY ((l1.l_partkey = (hashed SubPlan 6).col1) AND (l1.l_supplekey = (hashed SubPlan 6).col2)))) AND (ANY
Y (l1.l_orderkey = (hashed SubPlan 4).col1)))
      Rows Removed by Filter: 1343258
      Buffers: shared hit=1508 read=45476
      SubPlan 6
      -> Seq Scan on public.partsupp ps (cost=0.00..8272.00 rows=239976 width=8) (actual time=0.032..32.451 rows=240000.00 loops=1)
        Output: ps.ps_partkey, ps.ps_supplekey
        Filter: (ps.ps_avallqty > 0)
        Buffers: shared hit=376 read=4896
      SubPlan 4
      -> Seq Scan on public.orders o (cost=0.00..13521.00 rows=179865 width=4) (actual time=0.016..88.136 rows=180448.00 loops=1)
        Output: o.o_orderkey
        Filter: (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'::bpchar[]))
        Rows Removed by Filter: 269552
        Buffers: shared hit=756 read=7140
      -> Materialize (cost=0.00..72913.52 rows=4 width=30) (actual time=0.000..0.000 rows=0.00 loops=456835)
        Output: supplier.s_name, supplier.s_supplekey
        Storage: Memory Maximum Storage: 17kB
        Buffers: shared hit=70
      -> Nested Loop (cost=0.00..72913.50 rows=4 width=30) (actual time=0.981..0.983 rows=0.00 loops=1)
        Output: supplier.s_name, supplier.s_supplekey
        Join Filter: ((supplier.s_nationkey = nation.n_nationkey)
        Rows Removed by Join Filter: 610
        Buffers: shared hit=70
        -> Seq Scan on public.nation (cost=0.00..12.12 rows=1 width=4) (actual time=0.020..0.024 rows=1.00 loops=1)
          Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
          Filter: (nation.n_name = 'ARGENTINA'::bpchar)
          Rows Removed by Filter: 24
          Buffers: shared hit=1
        -> Seq Scan on public.supplier (cost=0.00..72892.00 rows=750 width=34) (actual time=0.076..0.929 rows=510.00 loops=1)
          Output: supplier.s_supplekey, supplier.s_name, supplier.s_address, supplier.s_nationkey, supplier.s_phone, supplier.s_acctbal, supplier.s_com
ment
          Filter: (((supplier.s_acctbal > '0'::numeric) AND (ANY (supplier.s_nationkey = (hashed SubPlan 2).col1)))
          Rows Removed by Filter: 2490
          Buffers: shared hit=69
          SubPlan 2
          -> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.034..0.040 rows=5.00 loops=1)
            Output: n.n_nationkey
            Hash Cond: (n.n_regionkey = r.r_regionkey)
            Buffers: shared hit=2
            -> Seq Scan on public.nation n (cost=0.00..11.70 rows=170 width=8) (actual time=0.002..0.004 rows=25.00 loops=1)
              Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment
              Buffers: shared hit=1
            -> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.017..0.018 rows=1.00 loops=1)
              Output: r.r_regionkey
              Buckets: 1024 Batches: 1 Memory Usage: 9kB
              Buffers: shared hit=1
              -> Seq Scan on public.region r (cost=0.00..12.12 rows=1 width=4) (actual time=0.010..0.011 rows=1.00 loops=1)
                Output: r.r_regionkey
                Filter: (r.r_name = 'EUROPE'::bpchar)
                Rows Removed by Filter: 4
                Buffers: shared hit=1
      -> Materialize (cost=0.00..43417498317.70 rows=150017 width=8) (never executed)
        Output: l3.l_orderkey, l3.l_supplekey
        -> Seq Scan on public.lineitem l3 (cost=0.00..43417496980.61 rows=150017 width=8) (never executed)
          Output: l3.l_orderkey, l3.l_supplekey
          Filter: (((l3.l_receiptdate > l3.l_commitdate) AND (ANY ((l3.l_partkey = (hashed SubPlan 14).col1) AND (l3.l_supplekey = (hashed SubPlan 14).col2)))) AND (ANY (l3.l_or
derkey = (hashed SubPlan 12).col1)))
          SubPlan 14
          -> Seq Scan on public.partsupp ps_2 (cost=0.00..8272.00 rows=239976 width=8) (never executed)
            Output: ps_2.ps_partkey, ps_2.ps_supplekey
            Filter: (ps_2.ps_avallqty > 0)
          SubPlan 12
          -> Seq Scan on public.orders o_2 (cost=0.00..13521.00 rows=179865 width=4) (never executed)
            Output: o_2.o_orderkey
            Filter: (o_2.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'::bpchar[]))
      -> Materialize (cost=0.00..43417496489.35 rows=450052 width=8) (never executed)
        Output: l2.l_orderkey, l2.l_supplekey
        -> Seq Scan on public.lineitem l2 (cost=0.00..43417492480.09 rows=450052 width=8) (never executed)
          Output: l2.l_orderkey, l2.l_supplekey
          Filter: (((ANY ((l2.l_partkey = (hashed SubPlan 10).col1) AND (l2.l_supplekey = (hashed SubPlan 10).col2)))) AND (ANY (l2.l_orderkey = (hashed SubPlan 8).col1)))
          SubPlan 10
          -> Seq Scan on public.partsupp ps_1 (cost=0.00..8272.00 rows=239976 width=8) (never executed)
            Output: ps_1.ps_partkey, ps_1.ps_supplekey
            Filter: (ps_1.ps_avallqty > 0)
          SubPlan 8
          -> Seq Scan on public.orders o_1 (cost=0.00..13521.00 rows=179865 width=4) (never executed)
            Output: o_1.o_orderkey
            Filter: (o_1.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'::bpchar[]))

Planning:
  Buffers: shared hit=255
  Planning Time: 28.668 ms
  Execution Time: 1247.121 ms
(121 rows)
```

=====

BENCHMARK SUMMARY

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```
Runs          : 5
Average Planning : 28.394 ms
Average Execution : 1233.670 ms
```

Generated Files:

```
run5_explain.txt
summary.txt
adityadave@Adityas-MacBook-Air-3 Q2&Q21 %
```